AI Inside YC-Startup Hubs and Shifting Narratives

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Abstract

We are interested in studying how the recent AI boom is reshaping the geography and thematic focus of Y Combinator (YC) startups. Using two complementary Hugging Face datasets, the YC companies directory (~4,000 startups) and the Hacker News archive (~40M posts). We plan to build an interactive Streamlit website that will allow users to explore where YC AI startups are located, how AI-related language in company descriptions evolves, and whether public discourse on Hacker News leads or follows YC portfolio shifts.

1. **How are YC AI startups geographically distributed, and has YC's Bay Area concentration changed over time, especially for AI companies vs. non-AI?**
2. **How has the language of YC company descriptions evolved around AI themes, and when did AI shift from niche to dominant category?**
3. **Does the rise of specific themes on Hacker News precede or coincide with those themes becoming prominent in YC cohorts?**

Note: Because the YC dataset has no explicit “AI” flag, we construct an AI label by combining industry/subindustry/tags with keyword rules applied to company descriptions, and validate it by manually labeling a random sample of 200 companies to report precision and recall.

Data Sources:

1.YC Companies-Hugging Face

Link: https://huggingface.co/datasets/datahiveai/ycombinator-companies

- Contents: One row per YC startup with name, one_liner, long_description, website, city, country, regions, industry, subindustry, tags, team_size, launched_at, batch, status, and several boolean flags.
- Use:
    - Define AI vs non-AI startups via industry/tags plus keyword rules on descriptions.
    - Analyze geographic distribution by city, country, and region across cohorts.
    - Run topic modeling and vocabulary analysis on descriptions to identify AI-related themes and track their evolution.

2. Hacker News-Hugging Face

Link: https://huggingface.co/datasets/open-index/hacker-news

- Contents: Stories, comments, Ask HN, Show HN, jobs, and polls since 2006, with text and timestamps.
- Use: We will filter to startup-relevant content (e.g., stories, Ask/Show HN) and use titles/text for topic trends around AI, crypto, devtools, fintech, and healthcare. We will construct simple yearly time series for a small set of themes and compare these to YC trends.

Planned Visualizations

Static visualizations

- Time-series line or stacked area charts showing the share of AI and major sectors in YC cohorts over time (by batch or year).
- Bar charts with top TF-IDF terms that distinguish AI vs non-AI YC descriptions and different AI sub-themes.
- Heatmaps of topic prevalence by year or batch for selected BERTopic topics derived from YC descriptions.

Interactive dashboard (Streamlit)

We will build a multi-page Streamlit app as the main public website:

- **Geography page (specialized: geospatial)**
  - Interactive maps (GeoPandas + Folium or PyDeck) of YC AI vs non-AI startups by city/country/region.
  - Filters for cohort range, AI flag, and potentially stage or status to compare early vs late cohorts and hub vs non-hub regions.
- **YC text themes page (specialized: text analysis)**
  - Topic and sector views: select an AI theme or sector and see its share over time as line/stacked area charts.
  - Tooltips or details-on-demand listing example YC companies for a chosen topic or sector.
- **Hacker News vs YC page**
  - For a small set of themes (AI, crypto, devtools, fintech), show time-series of frequencies in Hacker News vs YC descriptions.
  - Allow users to pick a theme and visually assess whether HN discussion tends to lead or follow YC portfolio shifts.

Methods and tools：

- **Data wrangling:** Pandas, NumPy.
- **Geospatial:** GeoPandas for spatial processing; Folium or PyDeck for mapping.
- **NLP & topics:** spaCy for tokenization/lemmatization; BERTopic or similar for topic modeling on YC descriptions (and possibly a filtered sample of Hacker News posts); scikit-learn TF-IDF for distinctive word analysis.
- **Visualization:** Altair or Plotly for static and interactive charts embedded in Streamlit.
- **Web app:** Streamlit for the interactive site, deployed on Streamlit Community Cloud.

Two expected hurdle:

- **AI labeling:** Since there is no built-in AI tag, we must rely on rules and small-sample manual checks, and we will document the uncertainty this introduces.
- **Hacker News scale:** The full archive is large, so we will limit ourselves to specific post types, a reasonable time window, and a small set of themes; if necessary, we can ship a fully complete project based on YC geography and YC text alone and treat the HN comparison as an extension.